Lab3
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Question 1
Play with your sorting hat. Are all 10 questions important to create the sorting hat? If you were to remove some questions to improve user experience, which questions would you remove and justify your answer.

Not all 10 questions are equally important for creating the sorting hat. Some questions overlap in the traits they measure, and a few seem less impactful in predicting house affiliation. For example, Question 3 (“Favorite subject?”) focuses more on academic preference than personality, which might not directly reflect house traits. Question 7 (“Preferred pet?”) is thematically relevant to Harry Potter but doesn’t contribute much to classification accuracy. Question 10 (“Dream career?”) is also more about future aspiration than current character. Removing these could improve the user experience by making the quiz shorter without significantly hurting model performance.

Question 2
If you were to improve the sorting hat, what technical improvements would you make? Consider:
- How could you improve the model's accuracy or efficiency?
- What additional sensors or hardware could enhance the user experience?
- Does decision tree remain suitable for your choice of new sensors? If yes, carefully justify your answer. If not, what ML model would you use and explain why.

To improve the sorting hat technically, several things can be done:
- To improve model accuracy, we could collect real user responses and apply better labeling rather than relying on strictly rule-based data. Feature selection methods could also help reduce noise and focus on the most relevant traits.
- For hardware, adding a speaker could enable audio narration of questions and results, making the interaction more engaging. Adding gesture control using an IMU sensor could replace button inputs for a more magical experience. RGB LEDs could be used to represent each house visually when the result is shown.
- The decision tree model is currently lightweight and efficient, which works well for an embedded system. However, if we introduce new sensors like gesture or voice input, we may need more expressive models. In that case, a random forest could still work while maintaining interpretability. Alternatively, a small neural network using TinyML would allow us to classify more complex input like time-series gesture data. For now, if we stay with button inputs and fixed options, the decision tree is still a suitable and effective model.